Unit 3

Solving systems of linear equations.

Unit 3.3

Gaussian Elimination with Augmented Matrices.

Linear systems, matrix notation, and augmented matrices

We can write a linear system as a set of individual equations:

$$\begin{array}{ccccc}
a_{11}x_1 & +a_{12}x_2 & +\cdots & +a_{1n}x_n & = b_1 \\
a_{21}x_1 & +a_{22}x_2 & +\cdots & +a_{2n}x_n & = b_2 \\
\vdots & \vdots & & \vdots & \vdots \\
a_{m1}x_1 & +a_{m2}x_2 & +\cdots & +a_{mn}x_n & = b_m
\end{array}$$

Linear systems, matrix notation, and augmented matrices

We can write the same system using matrix notation:

$$A\vec{x} = \vec{b}, \qquad A \in \mathbb{R}^{m \times n}, \vec{x} \in \mathbb{R}^{n}, \vec{b} \in \mathbb{R}^{m}$$

We can represent the same system as an augmented matrix:

$$M = \left[\begin{array}{c|c} A & \vec{b} \end{array}\right]$$

Just add *b* as an extra column to A.

Augmented matrices simplify work by hand.

The augmented matrix, $M = \left[\begin{array}{c|c} A & \vec{b} \end{array}\right]$, has all the same information as the original system, but we don't have to keep re-writing the variables (or any of the +/= signs).

$$M = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}, \qquad M \in \mathbb{R}^{m\times(n+1)}$$

This makes work by hand clearer and quicker.

This simplifies the mathematical representation in a computer.

The elimination process is identical!

Elementary row operations.

Both forward elimination and backward substitution fundamentally only involve three kinds of manipulation:

- **Addition of a scalar multiple**: Add a scalar multiple of one equation to another.

$$R_i + kR_j \rightarrow R_i$$

- **Scalar multiplication**: Multiply one equation by a scalar constant.

$$kR_i \rightarrow R_i$$

- **Row exchange**: Swap two equations.

$$R_i \leftrightarrow R_j$$

Using augmented matrices: an example.

Solve the system of equations:

$$\begin{array}{ccccccccc} x_1 & + & x_2 & + & x_3 & + & x_4 & = & 0 \\ x_1 & + & 2x_2 & + & 3x_3 & + & 4x_4 & = & 0 \\ 2x_1 & + & 2x_2 & + & 3x_3 & + & 3x_4 & = & 0 \\ 2x_1 & + & x_2 & + & 2x_3 & + & 1x_4 & = & 0 \end{array}$$

- First, recast as an augmented matrix:

$$M = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 4 & 0 \\ 2 & 2 & 3 & 3 & 0 \\ 2 & 1 & 2 & 1 & 0 \end{bmatrix}$$

Using augmented matrices: an example.

- Then proceed with Forward Elimination:

$$M=\begin{bmatrix}1&1&1&1&0\\1&2&3&4&0\\2&2&3&3&0\\2&1&2&1&0\end{bmatrix}\xrightarrow[R_4-2R_1\to R_4]{\substack{R_2-R_1\to R_2\\R_3-2R_1\to R_3}}\begin{bmatrix}1&1&1&1&0\\0&1&2&3&0\\0&0&1&1&0\\0&-1&0&-1&0\end{bmatrix}$$

$$\xrightarrow[R_4+R_2\to R_4]{R_3-0R_2\to R_3}\begin{bmatrix}1&1&1&1&0\\0&1&2&3&0\\0&0&1&1&0\\0&0&2&2&0\end{bmatrix}\xrightarrow{R_4-2R_3\to R_4}\begin{bmatrix}1&1&1&1&0\\0&1&2&3&0\\0&0&1&1&0\\0&0&0&0&0\end{bmatrix}=U$$

- We now have an **upper triangular** or **row echelon** form.

Back substitution with equations.

- Recast our upper triangular form as a system of equations:

$$U = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{array}{ccccccccc} x_1 & + & x_2 & + & x_3 & + & x_4 & = & 0 \\ & & x_2 & + & 2x_3 & + & 3x_4 & = & 0 \\ & & & & x_3 & + & x_4 & = & 0 \\ & & & & & & 0 & = & 0 \end{array}$$

- Note: We have a 0=0 row ~~than~~ that we can ignore, and one free variable, x_4.
- Now perform back substitution.

Back substitution with equations.

$$\begin{array}{ccccccccc} x_1 & + & x_2 & + & x_3 & + & x_4 & = & 0 \\ & & x_2 & + & 2x_3 & + & 3x_4 & = & 0 \\ & & & & x_3 & + & x_4 & = & 0 \end{array}$$

1. **Use x_3 as pivot to eliminate upwards:**
   - $R_2 - 2R_3 \to R_2$ **and** $R_1 - R_3 \to R_1$

$$\begin{array}{ccccccccc} x_1 & + & x_2 & & & & & = & 0 \\ & & x_2 & & & + & x_4 & = & 0 \\ & & & & x_3 & + & x_4 & = & 0 \end{array}$$

Back substitution with equations.

$$\begin{array}{ccccccccc} x_1 & + & x_2 & & & & & = & 0 \\ & & x_2 & & & + & x_4 & = & 0 \\ & & & & x_3 & + & x_4 & = & 0 \end{array}$$

2. **Use x_2 as pivot to eliminate upwards:**
   - $R_1 - R_2 \to R_1$

$$\begin{array}{ccccccc} x_1 & & & - & x_4 & = & 0 \\ & x_2 & & + & x_4 & = & 0 \\ & & x_3 & + & x_4 & = & 0 \end{array}$$

Back substitution with equations.

$$\begin{array}{ccccccccc} x_1 & & & & & - & x_4 & = & 0 \\ & & x_2 & & & + & x_4 & = & 0 \\ & & & & x_3 & + & x_4 & = & 0 \end{array}$$

3. Write all pivots (x_1, x_2, and x_3) in terms of the free variable x_4.

$$\begin{array}{ccccc} x_1 & & & = & x_4 \\ & x_2 & & = & -x_4 \\ & & x_3 & = & -x_4 \end{array}$$

4. Write final solution in parametric form:

$$(x_1, x_2, x_3, x_4) = (t, -t, -t, t), \forall t$$

Back substitution with augmented matrix.

- Our goal is to make each pivot entry the only non-zero entry in each column, and to make it equal to 1.
- This corresponds to isolating each free variable, just as we do in equation form.
- We start with the upper triangular form we got from forward elimination:

$$U = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Back substitution with augmented matrix.

$$U = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[R_2 - 2R_3 \to R_2]{R_1 - R_3 \to R_1} \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - R_2 \to R_1} \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = R$$

- We now have **reduced row echelon** or **row canonical** form.
  - All pivots are 1 and are the only non-zero entry in the corresponding column.

Back substitution with equations.

- Recast our reduced row echelon form as a system of equations:

$$U = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{array}{cccccccc} x_1 & & & - & x_4 & = & 0 \\ & x_2 & & + & x_4 & = & 0 \\ & & x_3 & + & x_4 & = & 0 \\ & & & & 0 & = & 0 \end{array}$$

- We can now write the solution:

$$\begin{array}{cccccc} x_1 & & & = & x_4 \\ & x_2 & & = & -x_4 \\ & & x_3 & = & -x_4 \end{array} \rightarrow (x_1, x_2, x_3, x_4) = (t, -t, -t, t), \forall t$$

Some notes on terminology

- The general result of forward elimination for any matrix is called a **row echelon** or **upper triangular** form of the matrix:
  - The first non-zero entry in every row has only zeros below and to the left of it.
- These leading entries on each row are the **pivots**.
- The general result of back substitution for any matrix is called a **reduced row echelon** or **row canonical** form of the matrix:
  - The pivot entries are the only non-zero entries in each column, the first non-zero entries in each row, and they are all equal to one.

Some notes on terminology

- **Gaussian elimination** is sometimes used to describe only the forward steps but may also describe the complete process.
- The complete forward elimination and back-substitution process is sometimes called **Gauss-Jordan elimination.**
- The process is also sometimes simply referred to as **row elimination**.

Forward elimination as an algorithm.

- Beginning with the augmented matrix, work from top to bottom and left to right.
- First, use the first non-zero entry in the top row to eliminate (zero out) all entries below it (this entry is called a pivot).
- Move to the second row and repeat.
- Move to the third row and repeat.
- …
- The result is a row echelon (upper triangular) form: pivots have zeros everywhere to the left and below.
- Note: We sometimes need to exchange two rows to give a non-zero pivot value we can work with.

Forward elimination as an algorithm.

Working with an $m \times (n+1)$ augmented matrix, $M = [A \quad b], A = [a_{ij}]$, start with row $i = 1$ and column $j = 1$, and repeat the following steps until $i = m$, or all rows below i contain only zeros:

- **If $a_{ij} = 0$ and $a_{kj} = 0$ for all rows below i ($a_{kj} = 0, \forall k \in [i, m]$), move to next column:**
$$j = j + 1$$
- **If $a_{ij} = 0$, but $\exists k > i$: $a_{kj} \neq 0$,, choose $k > i$ such that $a_{kj} \neq 0$, and exchange rows:**
$$R_i \leftrightarrow R_k$$
- **If $a_{ij} \neq 0$, use as pivot to eliminate column j for all rows below ($k \in [i+1, m]$) then move to next row and column:**
$$R_k - \left({}^{a_{kj}}/_{a_{ij}}\right) R_i \rightarrow R_k, \forall k \in [i+1, m], \qquad then\ j = j + 1\ and\ i = i + 1$$

Forward elimination checkpoints.

Considering the upper triangular form of an augmented matrix:

- If there are any rows that have a non-zero entry ONLY in the last column, the system has no solution; **do not proceed to back-substitution**.
  - These rows correspond to equations of the form $0 = b, b \neq 0$.
- If there are fewer non-zero rows than there are variables, then the system has many solutions; **back substitution is still needed to find the solutions**.
  - There must be columns, other than the last, that are not pivots; these are the free variables.
  - The final solution will express the pivot variables as functions of the free variables.
- If every column except the last contains a pivot, then the system has exactly one solution; **back substitution will give the solution directly**.

Back substitution as an algorithm.

- Beginning with the augmented matrix in upper triangular form, work from bottom to top and right to left.
- First, start with the first non-zero entry in the last non-zero row, using it to eliminate (zero out) all entries above it; divide this row by the value of the pivot so that the pivot entry is 1.
- Move to the second-to-last non-zero row and repeat.
- Move to the third-to-last non-zero row and repeat.
- …
- The result is a reduced row echelon (row canonical) form: pivots are all equal to one, and have zeros everywhere to the left, below and above.

Back substitution as an algorithm.

Working with an $m \times (n+1)$ augmented matrix in upper triangular form start with row $i = m$ and repeat the following steps until $i = 1$.

- **If row i is all zeros except for the last column ($a_{ij} = 0, \forall j \in [i, n]$ and $a_{i(n+1)} \neq 0$), there is no solution; STOP.**
- **If row i is all zeros ($a_{ij} = 0, \forall j \in [i, n+1]$), move up one row:**
$$i = i - 1$$
- **Otherwise, select j as the column with the first non-zero entry in row i and scale the row to make pivot a_{ij} equal 1. Then use a_{ij} (now equal 1) as pivot to eliminate column j for rows above ($k \in [1, i-1]$), and move up one row**
$$\left({}^{1}\!/_{a_{ij}}\right) R_i \rightarrow R_i,$$
$$then\;\; R_k - \left(a_{kj}\right) R_i \rightarrow R_k, \forall k \in [1, i-1],$$
$$then\;\; i = i - 1$$

Back substitution checkpoints.

Considering reduced row-echelon form of an augmented matrix:

- If every column except the last contains a pivot, then the system has exactly one solution; **the last column gives that solution directly**.
- If there are columns except the last that do not contain a pivot, then the system has multiple solutions; **translate back to equation form to easily write a general solution**.
  - The final solution will express the pivot variables as functions of the free variables.
- If there are any rows that have a non-zero entry ONLY in the last column, the system has no solution; **something has gone wrong if this wasn't the case in the upper triangular form given by forward elimination**.

Final comments.

- You may stop Gaussian elimination after the forward elimination steps and then use the algebraic equation form to back-substitute.
- However, performing the back-substitution steps in the augmented matrix form is often easier, and this approach can also be used to invert a matrix.
- While is it possible to solve any system of linear equations without exchanging rows, using “off-diagonal pivoting”, this should not be done.
- The connections to LU-decomposition, matrix inverses, and other fundamental concepts in linear algebra are lost when this approach is taken.

Unit 3.3.1

Worked examples solving linear systems using augmented matrices.

Big systems.

Solve the following system with 7 equations in 5 unknowns:

$$\begin{array}{cccccc}
x_1 & +x_2 & -x_3 & +2x_4 & +2x_5 & = -2 \\
 & x_2 & & +x_4 & -x_5 & = 1 \\
x_1 & +2x_2 & -x_3 & +3x_4 & -3x_5 & = -1 \\
 & x_2 & +x_3 & & & = 5 \\
x_1 & +x_2 & -x_3 & +3x_4 & -3x_5 & = -3 \\
 & & & x_4 & -x_5 & = -1 \\
x_1 & +2x_2 & -x_3 & +3x_4 & -2x_5 & = 4
\end{array}$$

Big systems.

- **First, recast in augmented matrix form:**

$$
\begin{array}{cccccc}
x_1 & +x_2 & -x_3 & +2x_4 & -2x_5 & = -2 \\
 & x_2 & & +x_4 & -x_5 & = 1 \\
x_1 & +2x_2 & -x_3 & +3x_4 & -3x_5 & = -1 \\
 & x_2 & +x_3 & & & = 5 \\
x_1 & +x_2 & -x_3 & +3x_4 & -3x_5 & = -3 \\
 & & & x_4 & -x_5 & = -1 \\
x_1 & +2x_2 & -x_3 & +3x_4 & -2x_5 & = 4
\end{array}
\rightarrow
\begin{bmatrix}
1 & 1 & -1 & 2 & -2 & -2 \\
0 & 1 & 0 & 1 & -1 & 1 \\
1 & 2 & -1 & 3 & -3 & -1 \\
0 & 1 & 1 & 0 & 0 & 5 \\
1 & 1 & -1 & 3 & -3 & -3 \\
0 & 0 & 0 & 1 & -1 & -1 \\
1 & 2 & -1 & 3 & -2 & 4
\end{bmatrix}
$$

Big systems.

- **Then, proceed with Gaussian Elimination:**

$$\begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 1 & 2 & -1 & 3 & -3 & -1 \\ 0 & 1 & 1 & 0 & 0 & 5 \\ 1 & 1 & -1 & 3 & -3 & -3 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 1 & 2 & -1 & 3 & -2 & 4 \end{bmatrix} \xrightarrow[R_7-R_1\to R_7]{\substack{R_3-R_1\to R_3 \\ R_5-R_1\to R_5}} \begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 1 & 0 & 1 & 0 & 6 \end{bmatrix}$$

Big systems.

- **Continue with Gaussian Elimination:**

$$\begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 1 & 0 & 1 & 0 & 6 \end{bmatrix} \xrightarrow{\substack{R_3 - R_2 \to R_3 \\ R_4 - R_2 \to R_4 \\ R_7 - R_2 \to R_7}} \begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 5 \end{bmatrix}$$

Big systems.

- **Continue with Gaussian Elimination:**

$$\begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 5 \end{bmatrix} \xrightarrow{R_3 to\ bottom} \begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Big systems.

- **Continue with Gaussian Elimination:**

$$\begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_5 - R_4 \to R_5} \begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Big systems.

- **Continue with Gaussian Elimination:**

$$\begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_5 to\ bottom} \begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- **Row echelon (upper triangular) form, with no free variables.**
- **There will be a single unique solution.**

Big systems.

- **Continue with Gaussian Elimination:**

$$\begin{bmatrix} 1 & 1 & -1 & 2 & -2 & -2 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[R_4+R_5\to R_4]{\substack{R_1+2R_5\to R_1 \\ R_2+R_5\to R_2 \\ R_3-R_5\to R_3}} \begin{bmatrix} 1 & 1 & -1 & 2 & 0 & 8 \\ 0 & 1 & 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Big systems.

- **Continue with Gaussian Elimination:**

$$\begin{bmatrix} 1 & 1 & -1 & 2 & 0 & 8 \\ 0 & 1 & 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[R_3+R_4\to R_3]{\substack{R_1-2R_4\to R_1 \\ R_2-R_4\to R_2}} \begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Big systems.

- **Continue with Gaussian Elimination:**

$$\begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1+R_3\to R_1} \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1-R_2\to R_1} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- **Reduced row echelon (row canonical) form, with no free variables.**
- **The last column contains the solution.**

Big systems.

- **Recast to equation form:**

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{matrix} x_1 = 1 \\ x_2 = 2 \\ x_3 = 3 \\ x_4 = 4 \\ x_5 = 5 \\ 0 = 0 \\ 0 = 0 \end{matrix}$$

- **In vector form:** $(x_1, x_2, x_3, x_4, x_5) = (1,2,3,4,5)$

Big systems.

Solve the following system with 7 equations in 5 unknowns:

$$\begin{array}{cccccc}
x_1 & +x_2 & -x_3 & +2x_4 & +2x_5 & = -2 \\
 & x_2 & & +x_4 & -x_5 & = 1 \\
x_1 & +2x_2 & -x_3 & +3x_4 & -3x_5 & = -1 \\
 & x_2 & +x_3 & & & = 5 \\
x_1 & +x_2 & -x_3 & +3x_4 & -3x_5 & = -3 \\
 & & & x_4 & -x_5 & = -1 \\
x_1 & +2x_2 & -x_3 & +3x_4 & -2x_5 & = 4
\end{array}$$

- **Solution:**

$$(x_1, x_2, x_3, x_4, x_5) = (1,2,3,4,5)$$

Multiple right-hand sides.

Solve the following system with two different right-hand sides:

Right-hand side 1:

$$\begin{array}{rrrrl} & & +x_3 & +x_4 & = 2 \\ 2x_1 & +x_2 & & +x_4 & = 3 \\ 2x_1 & +x_2 & +x_3 & +2x_4 & = 5 \\ 4x_1 & +2x_2 & +x_3 & +3x_4 & = 8 \end{array}$$

Right-hand side 2:

$$\begin{array}{rrrrl} & & +x_3 & +x_4 & = 3 \\ 2x_1 & +x_2 & & +x_4 & = 4 \\ 2x_1 & +x_2 & +x_3 & +2x_4 & = 7 \\ 4x_1 & +2x_2 & +x_3 & +3x_4 & = 12 \end{array}$$

Multiple right-hand sides.

RHS 1:
$$\begin{array}{ccccc} & & +x_3 & +x_4 & = 2 \\ 2x_1 & +x_2 & & +x_4 & = 3 \\ 2x_1 & +x_2 & +x_3 & +2x_4 & = 5 \\ 4x_1 & +2x_2 & +x_3 & +3x_4 & = 8 \end{array}$$

RHS 2:
$$\begin{array}{ccccc} & & +x_3 & +x_4 & = 3 \\ 2x_1 & +x_2 & & +x_4 & = 4 \\ 2x_1 & +x_2 & +x_3 & +2x_4 & = 7 \\ 4x_1 & +2x_2 & +x_3 & +3x_4 & = 12 \end{array}$$

- The rule of Gaussian elimination are **exactly the same** for these two systems, because all manipulations are chosen based on the variable coefficients.
- Augmented matrices gives us a trick to solve the systems at the same time!

$$A\vec{x} = \vec{b}_1 \text{ and } A\vec{x} = \vec{b}_2 \text{ lead to } M = \begin{bmatrix} A & \vec{b}_1 & \vec{b}_2 \end{bmatrix}$$

- Keep carefully in mind that we have two RHS columns.

Multiple right-hand sides.

RHS 1:
$$\begin{array}{lllll} & & +x_3 & +x_4 & = 2 \\ 2x_1 & +x_2 & & +x_4 & = 3 \\ 2x_1 & +x_2 & +x_3 & +2x_4 & = 5 \\ 4x_1 & +2x_2 & +x_3 & +3x_4 & = 8 \end{array}$$

RHS 2:
$$\begin{array}{lllll} & & +x_3 & +x_4 & = 3 \\ 2x_1 & +x_2 & & +x_4 & = 4 \\ 2x_1 & +x_2 & +x_3 & +2x_4 & = 7 \\ 4x_1 & +2x_2 & +x_3 & +3x_4 & = 12 \end{array}$$

- **Recast problem as an augmented matrix with two RHS columns:**

$$M = \begin{bmatrix} 0 & 0 & 1 & 1 & 2 & 3 \\ 2 & 1 & 0 & 1 & 3 & 4 \\ 2 & 1 & 1 & 2 & 5 & 7 \\ 4 & 2 & 1 & 3 & 8 & 12 \end{bmatrix}$$

- **Then proceed with Gaussian elimination.**

Multiple right-hand sides.

$$M = \begin{bmatrix} 0 & 0 & 1 & 1 & 2 & 3 \\ 2 & 1 & 0 & 1 & 3 & 4 \\ 2 & 1 & 1 & 2 & 5 & 7 \\ 4 & 2 & 1 & 3 & 8 & 12 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 2 & 1 & 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 5 & 7 \\ 4 & 2 & 1 & 3 & 8 & 12 \end{bmatrix}$$

$$\xrightarrow[R_4 - 2R_1 \to R_4]{R_3 - R_1 \to R_3} \begin{bmatrix} 2 & 1 & 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 1 & 2 & 3 \\ 0 & 0 & 1 & 1 & 2 & 3 \\ 0 & 0 & 1 & 1 & 2 & 4 \end{bmatrix} \xrightarrow[R_4 - R_2 \to R_4]{R_3 - R_2 \to R_3} \begin{bmatrix} 2 & 1 & 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Keep carefully in mind that we have two RHS columns.**

Multiple right-hand sides.

- Keep carefully in mind that we have two RHS columns.

$$\begin{bmatrix} 2 & 1 & 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 2 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 & 0 & 1 & 4 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- The first system has multiple solutions, while the second has none!
- Move forward with the first system only.

Multiple right-hand sides.

$$\begin{bmatrix} 2 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}R_1 \to R_1} \begin{bmatrix} 1 & {}^1\!/_2 & 0 & {}^1\!/_2 & {}^3\!/_2 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- x_2 and x_4 are free variables; solve for pivots x_1 and x_3 in terms of these.
- Recast to equation form to get the general solution:

$$\begin{array}{llll} x_1 + ({}^1\!/_2)x_2 & + ({}^1\!/_2)x_4 & = {}^3\!/_2 & \to \quad x_1 = {}^3\!/_2 - ({}^1\!/_2)x_2 - ({}^1\!/_2)x_4 \\ \qquad x_3 & + x_4 & = 2 & \quad\;\; x_3 = 2 - x_4 \end{array}$$

- For a parametric form, we set x_2=s and x_4=t:

$$(x_1, x_2, x_3, x_4) = \left({}^3\!/_2 - ({}^1\!/_2)s - ({}^1\!/_2)t, \quad s, \quad 2 - t, \quad t\right), \forall s, t \in \mathbb{R}$$

Multiple right-hand sides.

RHS 1:
$$\begin{array}{ccccc} & & +x_3 & +x_4 & =2 \\ 2x_1 & +x_2 & & +x_4 & =3 \\ 2x_1 & +x_2 & +x_3 & +2x_4 & =5 \\ 4x_1 & +2x_2 & +x_3 & +3x_4 & =8 \end{array}$$

RHS 2:
$$\begin{array}{ccccc} & & +x_3 & +x_4 & =3 \\ 2x_1 & +x_2 & & +x_4 & =4 \\ 2x_1 & +x_2 & +x_3 & +2x_4 & =7 \\ 4x_1 & +2x_2 & +x_3 & +3x_4 & =12 \end{array}$$

- **General solution for RHS 1:**

$$(x_1, x_2, x_3, x_4) = \left({}^3\!/_2 - \left({}^1\!/_2\right)s - \left({}^1\!/_2\right)t, \quad s, \quad 2 - t, \quad t\right), \forall s, t \in \mathbb{R}$$

- **RHS 2 has no solution.**

Computational Systems

In real applications of linear algebra, we almost always use computational tools to avoid needing to do the arithmetic by hand.

- Matlab (proprietary)
- Octave (Free and OpenSource)
- SciLab (Free and OpenSource)
    - All these work in quite similar ways, with Matlab and Octave being largely “syntax-compatible”
- It is helpful to get use to the basics of how to use these systems:
    - One easy way to get started is using “octave-online.net”

Unit 3.3.2

Worked examples solving real-world problems with elimination.

Energy production.

- A sustainable energy company is developing an electric power strategy for a new region; they will be deploying a combination of solar arrays, wind farms, and waste combustion plants.
- Each class of plant has different outputs that vary with season, and consumer demand also varies with season; all this is summarized in the following table (all values are in GWh/day):

	Spring	Summer	Fall	Winter
Solar	4	10	6	2
Wind	6	11	10	9
Waste	4	4	4	4
Demand	70	125	100	75

- **How many plants of each type should be deployed?**

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- How many plants of each type should be deployed?

- Let x_1 be the number of solar plants, x_2 be the number of wind farms, and x_3 be the number of waste combustion plants.
- Each season gives an equation, with daily energy production on the left-hand side and the daily demand on the right:
- Spring: $4x_1 + 6x_2 + 4x_3 = 70$
- Summer: $10x_1 + 11x_2 + 4x_3 = 125$
- Fall: $6x_1 + 10x_2 + 4x_3 = 100$
- Winter: $2x_1 + 9x_2 + 4x_3 = 75$
- We wish to solve this system for x_1, x_2, and x_3.

Energy production.

$$\begin{matrix} 4x_1 & +6x_2 & +4x_3 & = 70 \\ 10x_1 & +11x_2 & +4x_3 & = 125 \\ 6x_1 & +10x_2 & +4x_3 & = 100 \\ 2x_1 & +9x_2 & +4x_3 & = 75 \end{matrix} \to \begin{bmatrix} 4 & 6 & 4 & 70 \\ 10 & 11 & 4 & 125 \\ 6 & 10 & 4 & 100 \\ 2 & 9 & 4 & 75 \end{bmatrix} \xrightarrow{{}^{1}/_{2}R_1 \to R_1} \begin{bmatrix} 2 & 3 & 2 & 35 \\ 10 & 11 & 4 & 125 \\ 6 & 10 & 4 & 100 \\ 2 & 9 & 4 & 75 \end{bmatrix}$$

$$\xrightarrow[R_4 - R_1 \to R_4]{\substack{R_2 - 5R_1 \to R_2 \\ R_3 - 3R_1 \to R_3}} \begin{bmatrix} 2 & 3 & 2 & 35 \\ 0 & -4 & -6 & -50 \\ 0 & 1 & -2 & -5 \\ 0 & 6 & 2 & 40 \end{bmatrix} \xrightarrow{-{}^{1}/_{2}R_2 \to R_2} \begin{bmatrix} 2 & 3 & 2 & 35 \\ 0 & 2 & 3 & 25 \\ 0 & 1 & -2 & -5 \\ 0 & 6 & 2 & 40 \end{bmatrix} \xrightarrow{\substack{R_3 - {}^{1}/_{2}R_2 \to R_3 \\ R_4 - 3R_2 \to R_4}} \begin{bmatrix} 2 & 3 & 2 & 35 \\ 0 & 2 & 3 & 25 \\ 0 & 0 & -{}^{7}/_{2} & -{}^{35}/_{2} \\ 0 & 0 & -7 & -35 \end{bmatrix}$$

$$\xrightarrow{\substack{-{}^{2}/_{7}R_3 \to R_3 \\ -{}^{1}/_{7}R_4 \to R_4}} \begin{bmatrix} 2 & 3 & 2 & 35 \\ 0 & 2 & 3 & 25 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 1 & 5 \end{bmatrix} \xrightarrow{R_4 - R_3 \to R_4} \begin{bmatrix} 2 & 3 & 2 & 35 \\ 0 & 2 & 3 & 25 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

Energy production.

$$U=\begin{bmatrix}2&3&2&35\\0&2&3&25\\0&0&1&5\\0&0&0&0\end{bmatrix}\xrightarrow[R_2-3R_3\to R_2]{R_1-2R_3\to R_1}\begin{bmatrix}2&3&0&25\\0&2&0&10\\0&0&1&5\\0&0&0&0\end{bmatrix}\xrightarrow{{}^1\!/_2R_2\to R_2}\begin{bmatrix}2&3&0&25\\0&1&0&5\\0&0&1&5\\0&0&0&0\end{bmatrix}$$

$$\xrightarrow{R_1-3R_2\to R_1}\begin{bmatrix}2&0&0&10\\0&1&0&5\\0&0&1&5\\0&0&0&0\end{bmatrix}\xrightarrow{{}^1\!/_2R_1\to R_1}\begin{bmatrix}1&0&0&5\\0&1&0&5\\0&0&1&5\\0&0&0&0\end{bmatrix}=R\to\begin{array}{ccc}x_1&=&5\\x_2&=&5\\x_3&=&5\\0&=&0\end{array}$$

Energy production.

- A sustainable energy company is developing an electric power strategy for a new region; they will be deploying a combination of solar arrays, wind farms, and waste combustion plants.
- Each class of plant has different outputs that vary with season, and consumer demand also varies with season; all this is summarized in the following table (all values are in GWh/day):

	Spring	Summer	Fall	Winter
Solar	4	10	6	2
Wind	6	11	10	9
Waste	4	4	4	4
Demand	70	125	100	75

- **Deploying 5 plants of each type would perfectly satisfy the demands in all four seasons.**

Economic equilibrium.

- A coarse model of a national economy breaks economic activity into three sectors: primary (agriculture and natural resource extraction); secondary (manufacturing); and tertiary (services).
- Consider a “balanced” economy, where to the production of $1M of output from any sector requires $0.3M of input from each of the three sectors.
- If there is an additional (e.g. consumer) demand for $100M from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.

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- Let x_i be the total supply (in millions of dollars produced) by economic sector *i*.
- Let d_i be the total demand (in millions of dollars needed) for the output of economic sector *i*.
- Let the sectors be defined as: agriculture & natural resources *i=1*; manufacturing *i=2*; and services *i=3*.

$$\begin{aligned} d_1 &= 0.3x_1 + 0.3x_2 + 0.3x_3 + 100 \\ d_2 &= 0.3x_1 + 0.3x_2 + 0.3x_3 + 100 \\ d_3 &= 0.3x_1 + 0.3x_2 + 0.3x_3 + 100 \end{aligned}$$

- We would like to find a consistent solution, where the total supply is equal to the total demand for each sector of the economy.

$$d_i = x_i, \forall i$$

Economic equilibrium.

$$\begin{aligned} x_1 &= 0.3x_1 &+0.3x_2 &+0.3x_3 &+100 \\ x_2 &= 0.3x_1 &+0.3x_2 &+0.3x_3 &+100 \\ x_3 &= 0.3x_1 &+0.3x_2 &+0.3x_3 &+100 \end{aligned}$$

$$\vec{x} = A\vec{x} + \vec{b}, A = \begin{bmatrix} 0.3 & 0.3 & 0.3 \\ 0.3 & 0.3 & 0.3 \\ 0.3 & 0.3 & 0.3 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 100 \\ 100 \\ 100 \end{bmatrix}$$

$$\vec{x} = A\vec{x} + \vec{b} \rightarrow \vec{x} - A\vec{x} = \vec{b} = (I - A)\vec{x} = \vec{b}$$

$$M\vec{x} = \vec{b}, M = \begin{bmatrix} +0.7 & -0.3 & -0.3 \\ -0.3 & +0.7 & -0.3 \\ -0.3 & -0.3 & +0.7 \end{bmatrix} \rightarrow \begin{bmatrix} +0.7 & -0.3 & -0.3 & 100 \\ -0.3 & +0.7 & -0.3 & 100 \\ -0.3 & -0.3 & +0.7 & 100 \end{bmatrix}$$

Economic equilibrium.

$$\begin{bmatrix} +0.7 & -0.3 & -0.3 & 100 \\ -0.3 & +0.7 & -0.3 & 100 \\ -0.3 & -0.3 & +0.7 & 100 \end{bmatrix} \xrightarrow[10R_3 \to R_3]{\substack{10R_1 \to R_1 \\ 10R_2 \to R_2}} \begin{bmatrix} +7 & -3 & -3 & 1000 \\ -3 & +7 & -3 & 1000 \\ -3 & -3 & +7 & 1000 \end{bmatrix}$$

$$\xrightarrow{\substack{R_2 + {}^3/_7 R_1 \to R_2 \\ R_3 + {}^3/_7 R_1 \to R_3}} \begin{bmatrix} 7 & -3 & -3 & 1000 \\ 0 & \left(7 - \frac{9}{7}\right) & \left(-3 - \frac{9}{7}\right) & \left(1000 + \frac{3000}{7}\right) \\ 0 & \left(-3 - \frac{9}{7}\right) & \left(7 - \frac{9}{7}\right) & \left(1000 + \frac{3000}{7}\right) \end{bmatrix} = \begin{bmatrix} 7 & -3 & -3 & 1000 \\ 0 & \frac{40}{7} & \frac{-30}{7} & \frac{10000}{7} \\ 0 & \frac{-30}{7} & \frac{40}{7} & \frac{10000}{7} \end{bmatrix}$$

$$\xrightarrow{\substack{{}^7/_{10} R_2 \to R_2 \\ {}^7/_{10} R_3 \to R_3}} \begin{bmatrix} 7 & -3 & -3 & 1000 \\ 0 & 4 & -3 & 1000 \\ 0 & -3 & 4 & 1000 \end{bmatrix} \xrightarrow{R_3 + {}^3/_4 R_2 \to R_3} \begin{bmatrix} 7 & -3 & -3 & 1000 \\ 0 & 4 & -3 & 1000 \\ 0 & 0 & \left(\frac{7}{4}\right) & 1750 \end{bmatrix} = U$$

Economic equilibrium.

$$U = \begin{bmatrix} 7 & -3 & -3 & 1000 \\ 0 & 4 & -3 & 1000 \\ 0 & 0 & \left(\dfrac{7}{4}\right) & 1750 \end{bmatrix} \xrightarrow{4/_7 R_3 \to R_3} \begin{bmatrix} 7 & -3 & -3 & 1000 \\ 0 & 4 & -3 & 1000 \\ 0 & 0 & 1 & 1000 \end{bmatrix} \xrightarrow[R_2+3R_3 \to R_2]{R_1+3R_3 \to R_1} \begin{bmatrix} 7 & -3 & 0 & 4000 \\ 0 & 4 & 0 & 4000 \\ 0 & 0 & 1 & 1000 \end{bmatrix}$$

$$\xrightarrow{1/_4 R_2 \to R_2} \begin{bmatrix} 7 & -3 & 0 & 4000 \\ 0 & 1 & 0 & 1000 \\ 0 & 0 & 1 & 1000 \end{bmatrix} \xrightarrow{R_1+3R_2 \to R_1} \begin{bmatrix} 7 & 0 & 0 & 7000 \\ 0 & 1 & 0 & 1000 \\ 0 & 0 & 1 & 1000 \end{bmatrix} \xrightarrow{1/_7 R_1 \to R_1} \begin{bmatrix} 1 & 0 & 0 & 1000 \\ 0 & 1 & 0 & 1000 \\ 0 & 0 & 1 & 1000 \end{bmatrix} = R$$

$$\begin{aligned} x_1 &= 1000 \\ \to x_2 &= 1000 \\ x_3 &= 1000 \end{aligned}$$

Economic equilibrium.

- A coarse model of a national economy breaks economic activity into three sectors: primary (agriculture and natural resource extraction); secondary (manufacturing); and tertiary (services).
- Consider a “balanced” economy, where to the production of \$1M of output from any sector requires \$0.3M of input from each of the three sectors.
- If there is an additional (e.g. consumer) demand for \$100M from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.
- Each sector must product \$1000M, or \$1B.
- The consumer demand represents only 10% of the total demand for each sector.